

REQUEST FOR INFORMATION



Safety Assured Control and Security Architecture for Mission Autonomy

RFI Number: RCI-2026-002

Issue Date: March 11, 2026
Response Deadline: April 9, 2026

1. Executive Summary

The Army's Capability Program Executive Mission Autonomy (CPE MA) PM Robotic Control and Integration (PM RCI) is seeking scalable modular industry hardware and software solutions against a common architecture that delivers safety assured control for uncrewed air, ground and maritime platforms and integrated lethal and non-lethal payloads for mission autonomy. Input and submittals are requested from small businesses, non-traditional firms and large business concerns, and CPE MA is seeking complete or partial solutions and/or niche concepts or products.

This document serves as a Request for Information (RFI)/Sources Sought for technologies which show potential for meeting the description below. Technologies and solutions may be Commercial Off-the-Shelf (COTS) or developmental items. This is not a Request for Quotation (RFQ), Request for Proposal (RFP) or an Invitation for Bid (IFB), but submittals may form the genesis of future CPE MA requirements. This RFI is a market research tool being used to determine the availability and adequacy of potential sources for planning purposes only. Responses to this notice are not considered as offers and cannot be accepted by the Government to form a binding contract. The Government does not intend to award a contract on the basis of responses nor otherwise pay for the preparation of any information presented or the use of any such information. It is plausible that the Government may later issue an RFWP/RFP, depending on mission requirements; thus, submittals must state whether the Company/Entity would likely participate in future proposals for the concepts outlined in their technical write up.

2. Purpose

CPE MA is conducting market research/sources sought to identify industry capabilities for software and hardware solutions to provide a safety assured control and security architecture for uncrewed air, ground and maritime platforms and their integrated lethal and non-lethal payloads operating under remote control and/or autonomous control. The goal is to provide an independent verifiable level of assured control with safety-critical functions of robotic autonomous systems under various levels of supervision. Supervision may range from an operator in-the-loop to on-the-loop, to complete local autonomy, all in a variety of operational domains.

3. Autonomy Definitions

Platform Autonomy: Platform autonomy is focused on mobility and maneuver in the air, on and below the ground, and on or below the surface of water. This encompasses the autonomous capabilities required for navigating air and ground platforms and interacting with the physical environment, such as spatial understanding, obstacle avoidance, and path optimization. Platform autonomy also enables the system to maneuver tactically through the environment to a position of relative advantage, considering signature management, sight lines, and known or suspected enemy positions.

Payload Autonomy: Payload autonomy is centered on the effects and functionality of the specific payloads carried by an autonomous system. This includes the integration and operation of mission-specific equipment, such as sensors, weapons, communication devices, or other

specialized packages. Payload autonomy allows the system to independently utilize its payloads to achieve desired effects without requiring constant human oversight.

Mission Autonomy: This aspect of autonomy primarily focuses on planning and collaboration and is an integration layer above and between payload autonomy and platform autonomy. It involves the systems and algorithms responsible for high-level decision-making, such as route planning, task prioritization, and mission execution. Mission autonomy enables an autonomous system to operate cohesively with other units, whether human or machine, by interpreting mission objectives, generating plans, and adjusting those plans dynamically based on changes in the environment or mission priorities.

Collaboration is a key feature of mission autonomy, as it requires robust communication protocols and the ability to synchronize actions across a network of systems. This ensures that the autonomous system components contribute effectively to the larger operational framework, whether in isolation or as part of a team. Control and the associated user interface are critical components of mission autonomy and must be designed to include human factors considerations and to enable the objective of one-to-many control of heterogenous autonomous systems.

4. Assured Control Safety Architecture

An assured safety architecture for autonomous vehicle and payload operations is a unified, verifiable framework that maps the safe execution of a mission from start to finish. It manages and synergizes the safety requirements of uncrewed platforms and integrated mission systems. An example safety assured control and security architecture and its subordinate modules are illustrated below.

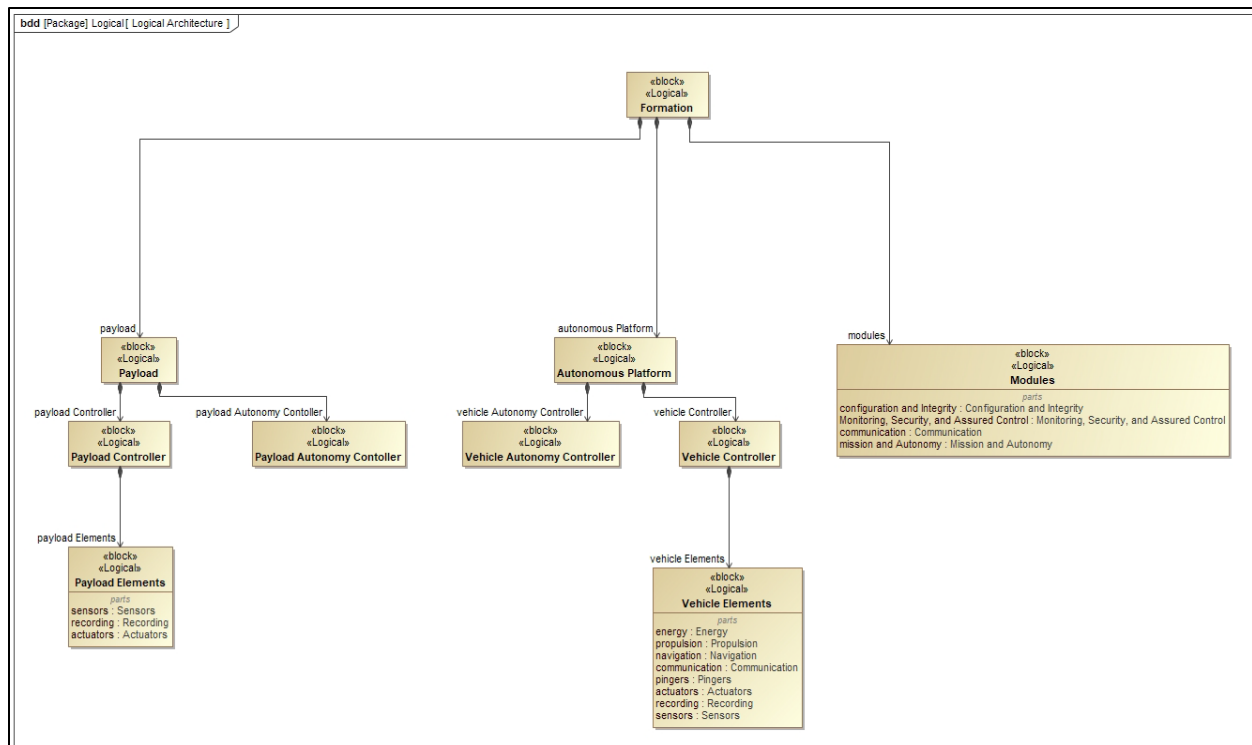


Figure 1: Example Autonomy Logical Architecture

Figure 1 shows an example autonomy logical architecture intended to illustrate a means of achieving assured control; it is not a definitive design. A formation consists of autonomous platforms with optional payloads and modules. The modules provide safety and cybersecurity enablers that form a basis for assured control. Figure 2 shows the modules in a hierarchy to summarize what they assure: the mission, communications, unit health and control, or unit configuration.

These modules are intended to be optional and combinable in various ways. Figure 3 provides one possible composition as a context diagram. This approach allows a unit to contain only the hardware and software required to accomplish its mission, preserving flexibility and optimizing cost.

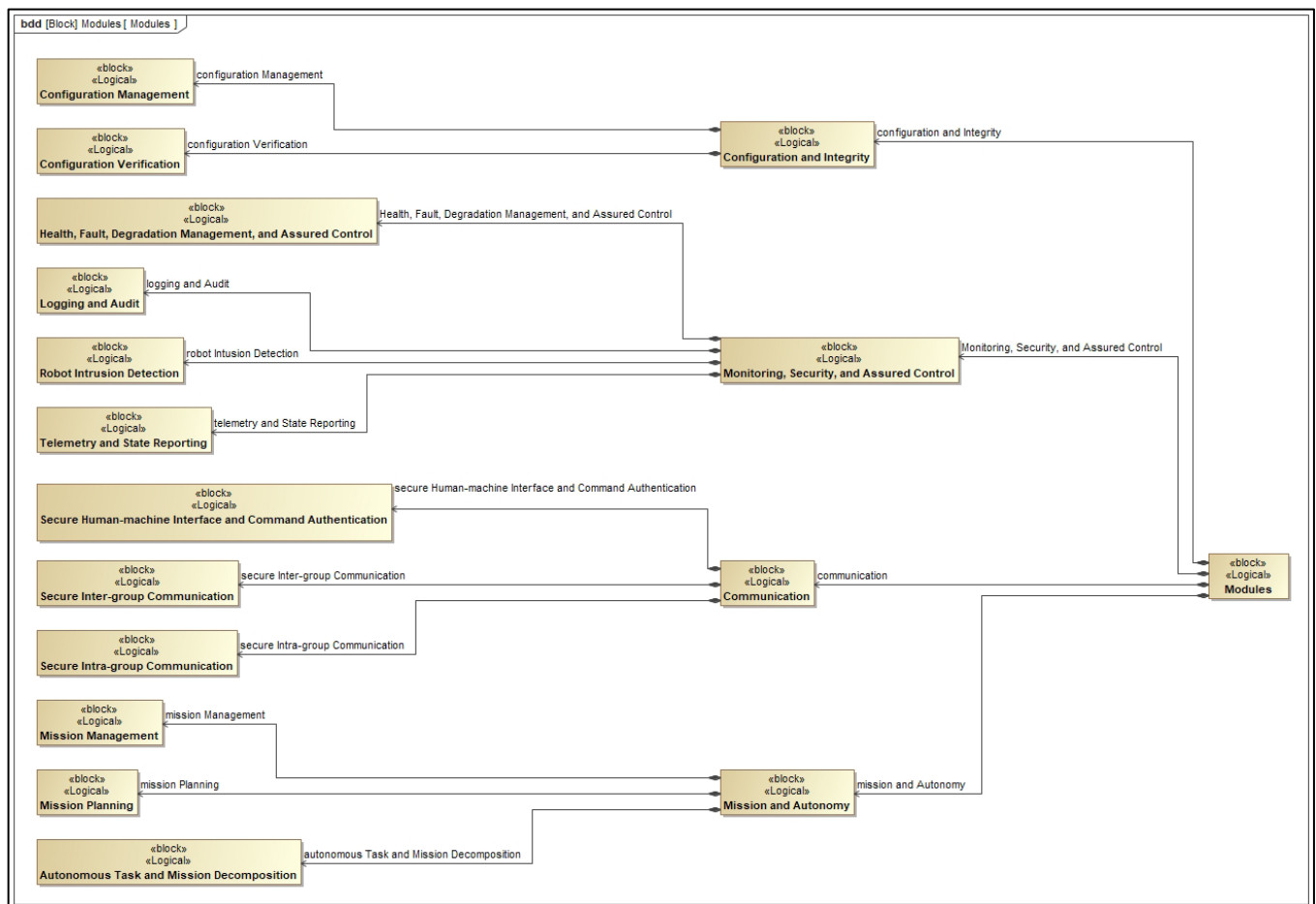


Figure 2: Example Autonomy Logical Architecture Modules

The example context diagram (Figure 2) depicts 12 modules that implement:

- Configuration management and verification to cryptographically assure that units are composed of the correct hardware and software with the expected settings
- Assured control through health, fault, and degradation monitoring with flexible interventions

- Operational, mission, and command transparency through telemetry / state collection and logging / audit
- Intrusion detection
- Secure human-machine interaction and command authentication
- Secure and robust communication within and between groups
- Mission management to allow safe and secure mission delivery, updates, and aborts
- Mission planning with safety and cybersecurity verification
- Mission decomposition into tasks delivered to payloads or other units

These modules implement the core functions and key attributes discussed in the following sections.

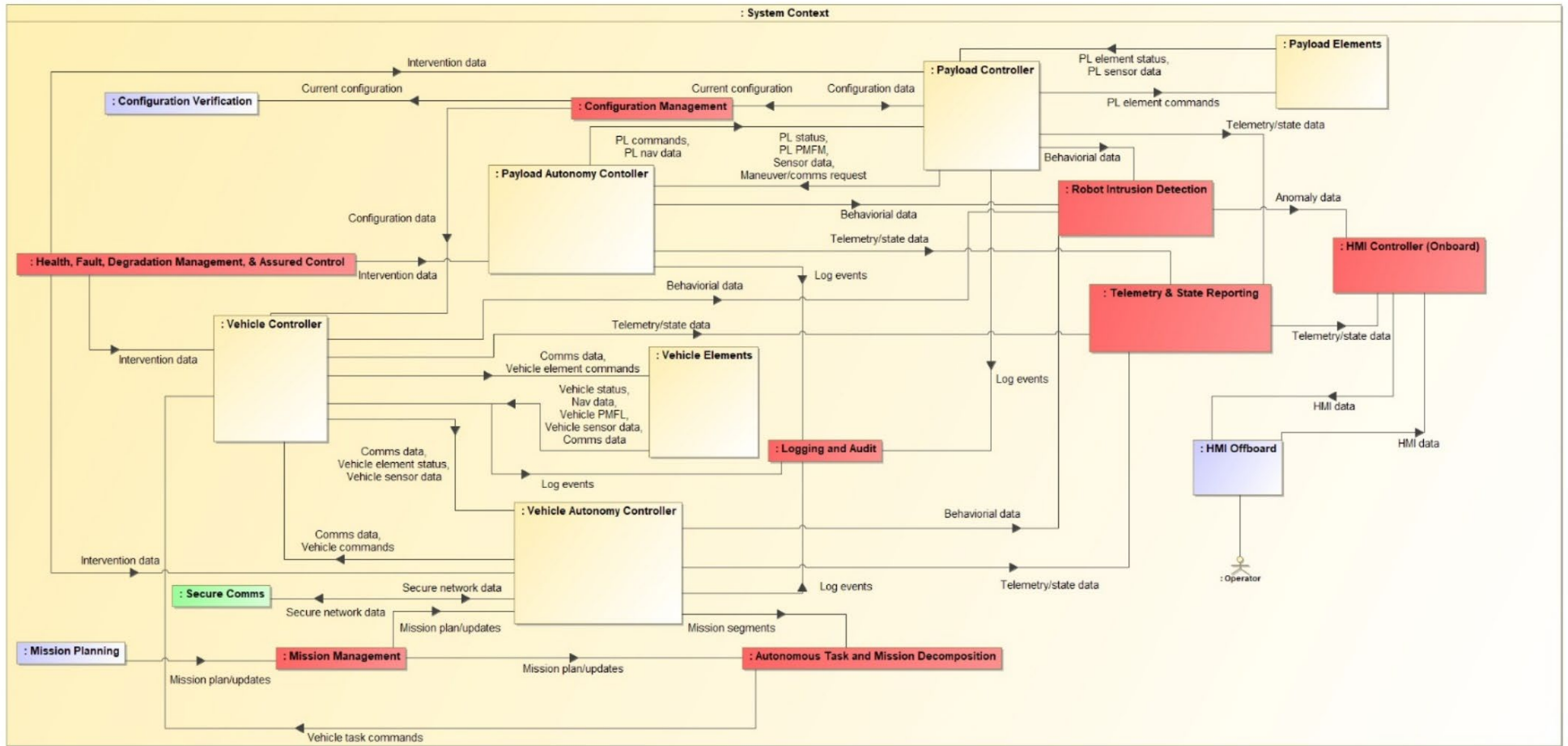


Figure 3: Context Diagram

5. Core Functions

A robust safety architecture for mission autonomy incorporates critical functions of a platform safety system, governs payload operations and orchestrates the mission in complex environments, varying tactical domains and scenarios.

Function	Description
Holistic State Manager	Continuously tracks the status of both the vehicle (e.g., position, speed, battery state) and the payload (e.g., armed/safe, stowed/deployed, active/inactive). It must include the allowable transitions between states.
Fault Management & Mitigation	Responds to failures in <i>both</i> the platform and payload subsystems. It must then execute a Coordinated Minimal Risk Condition (MRC). This may mean simply stopping the vehicle, but it may also require more complex behaviors such as safely stowing the payload, deactivating a weapon system, or other action <i>before</i> the vehicle performs its own safety maneuver.
Dynamic Safety Envelope Management	Manages a multi-dimensional "safety bubble" that changes based on the combined state of the platform(s) and payload(s), and the tactical domain and context of the mission.
Safe Activity Sequencing	Enforces a verified sequence of operations. For example, it may prevent a platform from attempting to move at high speed while in a high risk terrain or tactical maneuver, or it may ensure a weapon payload can only be armed after a series of specific safety checks are verified.
Verification & Validation (V&V)	Provides the framework for rigorously testing and validating remote control and/or autonomous commands. The V&V process must prove that state transition and payload action is safe and that failure modes are handled gracefully by the integrated system.

6. Key Attributes

The attributes of the architecture must consider the autonomous platform, its mission and the added complexity and risk of payload operations.

Attribute	Description
Redundancy & Diversity	This applies to both the platform and critical payload systems. For instance, a payload might have redundant actuators or safety interlocks. Diversity is key, such as using parallel sensors or processors to identify control inconsistencies and system failures so that consequences can be mitigated prior to a larger-scale failure.
Segregation & Partitioning	The architecture must create a virtual "firewall" between flight-critical/drive-critical control systems and the payload/mission systems. A software crash in the payload's user interface must not be able to affect the vehicle's ability to drive or fly safely. This is often achieved through specialized real-time operating systems and hardware partitioning.

Hierarchical Safety Control	The architecture may be layered. A low-level system ensures basic platform functionality and stability. A higher-level "Safety Arbiter" monitors both a platform and payload commands and will veto any action—from either system—that would violate the overall mission safety envelope in a given domain.
Data and Command Integrity	The architecture must ensure that commands sent to the payload (e.g., "activate sensor," "release package," "fire weapon") are coming across a secure channel and that the commands have been received correctly. This prevents both accidental activation and malicious attacks.
Comprehensive Safety Case	The documentation for the system must provide a clear, evidence-backed argument for why the integrated system is safe. This safety case must explicitly address the risks of the payload(s), the platform(s), and their interactions in a mission context, and demonstrate how they are mitigated by the architecture.

7. Information Requested

The Government is interested in proven or emerging alternatives to the proposed assured control and security architecture that are open source and include solutions built to the offered architecture. Please provide a capabilities statement addressing the following:

1. **Architecture Description and Model:** Provide a description of the proposed assured control safety and security architecture, its hardware and software modules and how they collectively deliver an assured control solution for mission autonomy.
2. **Integration Approach:** Describe how your hardware and software integration handles data exchange with existing safety-critical modules, system control and autonomous software for supervised and non-supervised missions, Assured Armaments Reference Architecture (AARA) and adherence to Modular Open Systems Approach (MOSA) disciplines and how it ensures interoperability.
3. **Wireless Transport Approach:** Explain how your architecture integrates with wireless transport media.
4. **Safety Compliance:** Detail your experience developing software to Automotive Safety Integrity Level D (ASIL-D) or equivalent rigorous safety standards, e.g. MIL-STD-882E.
5. **Security Compliance:** Detail your experience developing software to meet DoW or industrial security standards.
6. **Command Arbitration:** Explain your methodology for prioritizing, combining and validating signals from multiple sources (manual vs. autonomous).
7. **Technical Maturity:** Provide examples of where your safety assured control architecture has been implemented/integrated. Describe the level of Safety Verification/Validation that your company's solution has undergone. Evidence of performance is highly encouraged.

8. Describe what test cases you have developed to test your safety architecture solutions. Describe key metrics used in such testing and verification.

8. RFI Response Instructions

Responses should be submitted electronically to the point of contact listed in Section 10. All submissions must be received by April 9, 2026, at 5:00 PM Eastern Time.

Page Limit and Format

Technical response: Maximum 15 pages (excluding cover page and appendices)

Font: Arial or Times New Roman, 11-point minimum

Format: PDF, Microsoft Word, or equivalent

Margins: 1 inch on all sides

The following additional information is requested in your response:

- A rough schedule of tasks, major milestones leading to the delivery of a prototype or production ready solution.
- Describe your company's solution control software's ability to be operated from third-party computing hardware, display, and physical control interfaces.
- Describe how your solution's autonomous behaviors are exportable to other platforms/payloads. What standards are needed to be followed for such exportation?
- Provide a rough order of magnitude (ROM) cost/price based on the planned schedule and tasks.

Cover Page: Submittals must include a Company or Entity cover page, limited to one (1) page, which will not count against the 15-page limit, and should include Company name and address, and CAGE Code (if available); the Technical and Business Points of Contact (POC) to include: Telephone Numbers and Email Addresses; Company headquarters location; size of company, both revenue and # of employees; if the Company is designated as a non-traditional or traditional contractor; if the Company's proposed solution/approach has been proposed to or previously funded by any federal agency, to include SBIR/STTR, please identify each specific agency and all applicable contract/agreement numbers; and a brief abstract (1,000 character limit which should be included in the appendix). Also state whether you are a member of NAMC or any other defense related consortium. The technical submittals should include the contribution to the requirement and provide a high-level project overview describing how this contribution addresses the stated requirements.

Classification: This RFI and all responses shall be UNCLASSIFIED. Do not include any classified, proprietary, or export-controlled information unless specifically requested and cleared through appropriate channels.

9. Important Notices and Disclaimers

9.1. Nature of this RFI

This RFI is issued for information and planning purposes only and does not constitute a Request for Proposal (RFP) or a promise to issue an RFP. This RFI does not commit the Government to

contract for any supplies or services. Responses to this RFI are not offers and cannot be accepted by the Government to form a binding contract.

9.2. No Funding for Responses

Respondents are solely responsible for all expenses associated with responding to this RFI. The Government will not reimburse any costs incurred in preparing or submitting responses.

9.3. Proprietary Information

Respondents should clearly mark any proprietary or business-sensitive information. The Government acknowledges its obligations under the Federal Acquisition Regulation (18 U.S.C. §1905) to protect confidential information provided to the USG. Pursuant to this statute, the Government is willing to accept any trade secrets or proprietary restrictions placed upon data presented in response to this RFI and to protect such information from unauthorized disclosure. While the Government will make reasonable efforts to protect such information, responses may be subject to the Freedom of Information Act (FOIA). Respondents should not include information they wish to protect from public disclosure unless absolutely necessary.

9.4. Government Rights/Disclaimer

The USG is not responsible for the public disclosure of unmarked data received in response to this RFI. Accordingly, the potential offeror shall ensure that technical data provided in response to this RFI shall be appropriately marked (i.e., PROPRIETARY) to ensure proper Government handling of the data. Additionally, marked data that is already in the public domain or in the possession of the Government or third parties, or is afterward placed into the public domain by the owner or another party through no fault of the Government will not be protected once in the public domain. Data already in the possession of the Government will be protected in accordance with the USG's rights in the data. Should the Government need to reproduce the protected data for distribution purposes between Government offices, all such data shall be reproduced with restrictive legends in place. The Government may employ AI as a tool to assist in the analysis and review of RFI submissions. AI tools may be used to support tasks such as summarizing content, and highlighting areas of potential strength, weakness, or risk. The Government retains sole responsibility for all inherently governmental functions and will not delegate decision making authority to any AI tool. The AI tools utilized by the Government will be employed in a manner consistent with applicable regulations that govern the protection of proprietary and source selection sensitive information. The AI tool will be an output-only system. Any generated report data is not saved. No RFI data will be transmitted to or processed by external, non-Government systems without appropriate safeguards and authorizations in place. USG support contractors may assist in reviewing RFI submissions. Information submitted in response to this RFI may be used by the Government for any purpose, including but not limited to: developing requirements, preparing solicitations, evaluating proposals, or sharing with other Government agencies. The Government may discuss information in responses with other potential contractors as part of market research activities.

9.5. Questions

If a vendor has questions, they shall be included as an appendix to the vendor's RFI response.

9.6. Related RFIs

CPE MA has released the related “Robotic Ground Combat Systems RFI” with NAMC via <https://www.namconsortium.org/opportunities/rfi-00-27-cpe-mission-autonomy> and “Landscape of Current Industry Robotic Solutions,” with SAM.GOV via <https://sam.gov/workspace/contract/opp/f37847b6c34e42bdb4c2f9f835649594/view>. An RFI entitled “Multi-Entity Autonomy-Enabled Mission Command Reference Architecture” is anticipated in the near future.

10. RFI Submission Instructions

All inquiries and responses should be directed to:

General questions to: questions@namconsortium.org

RFI Response Submissions to: submissions@namconsortium.org

Subject Line: "RFI Response - RCI-2026-002 - [Company Name]"

RFI Response Deadline - April 9, 2026, at 5:00 PM ET

Maximum file size: 20 MB (Reach out to NAMC via questions@namconsortium.org if a larger submission is required)

A panel of Army and contractor SMEs will review submittals. It is expected that discussions may occur between Army SMEs and the contractor/performer for the purpose of clarifying technical submissions. All RFI submissions are treated as company proprietary information, and the content will be disclosed to U.S. Government (USG) employees, military, or designated support contractors only for the purpose of reviewing technical feasibility and market research. RFI submittals should outline both capacities and technical solutions/concepts and prototypes that address and support the stated objectives and requirements outline herein. The 15-page technical submittal, in contractor format, may focus on describing details of a proposed solution or prototype, how it could address the technical requirements, and potential impacts and benefits to the Army’s Autonomous Mission.

11. Key Dates

RFI Release Date - March 11, 2026

RFI Response Deadline - April 9, 2026, at 5:00 PM ET

Note: All times are Eastern Time (ET). Late submissions will not be accepted unless explicitly authorized by the Government.